

Quantum Gravitation

Problem Set 9

Yuran Seo

Due 2026-11-01 23:59

Topic

Renormalization and Parameter Matching

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Renormalization and Parameter Matching. State all assumptions and conventions.

$$g_i^0 = \mu^{d_i \epsilon} (g_i + \delta g_i)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\beta_i = \mu dg_i/d\mu$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$g_i(\mu_{obs}) = U_{ij} g_j(\mu_{calc})$$

Show enough intermediate work that another student can reproduce the calculation.